

KOLLI CHARAN ADITHYA

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EDUCATION

Bachelor of Technology in Computer Science and Artificial Intelligence

2023 – Present

Newton School of Technology, Rishihood University

CGPA: 7.24 / 10.0

EXPERIENCE

Product Engineer Intern

Nov 2024 – Apr 2025

IIT Roorkee (SSF)

Remote — Next.js, TypeScript, REST APIs, Cypress, Figma

- Reduced initial bundle size and improved load times on low-bandwidth networks by implementing **code-splitting and lazy-loading** in a React portal serving government schemes, scholarships, and job listings.
- Lifted application completion rate by **40%** by designing and shipping a **recommendation engine** that personalizes scheme/job matching using user profile data and behavioral signals.
- Cut production regressions by building a **Python scraping pipeline** to ingest & normalize scheme data from government portals, and authoring **50+ Cypress E2E tests** covering critical user flows.
- Designed a mobile-first UI/UX system in Figma (0-to-1 screens) and translated it into a reusable React component library adopted platform-wide for **260M+ users**.

PROJECTS

DraftForge (Demo) (GitHub)

Next.js, TypeScript, Tailwind CSS, Supabase, PostgreSQL, WebSockets, Postgres CDC

- Built a concurrency-safe **bidding engine** (PostgreSQL stored procedure + `SELECT ... FOR UPDATE`) that serializes bids across **10 concurrent players** while enforcing purse and quota rules — zero race conditions.
- Designed an **idempotent auction state-machine** with optimistic concurrency guards that prevents double-advancement on timer expiry using a staggered host/peer trigger protocol.
- Architected real-time sync across **5 Postgres tables** via CDC over filtered WebSocket channels (Presence + Broadcast) for consistent state across **10 concurrent clients**.

ModelRoute — LLM Routing Engine & API Gateway (Demo) (GitHub)

TypeScript, Next.js 15, PostgreSQL, Upstash Redis, Groq, Gemini, Ollama

- Built a high-throughput **API gateway + LLM routing engine** that classifies prompt intent in sub-millisecond regex across **8 task categories** and dispatches to the optimal provider (Groq / Gemini / Ollama) by latency, cost, and SLA.
- Implemented a state-machine **Circuit Breaker** with exponential backoff for zero-downtime failover, plus **SHA-256 key authentication** and Redis sliding-window rate limiting.
- Delivered real-time token streaming via **Server-Sent Events (SSE)** and enforced strict runtime schema checks at every boundary with a **Zod-validated REST API** layer.

SKILLS

Languages	TypeScript, JavaScript, Python, SQL
Frameworks	React, Next.js, Node.js, Express.js, Tailwind CSS
Databases & APIs	PostgreSQL, Prisma, Supabase, REST APIs, WebSockets, JWT/OAuth
Tools	Git, GitHub, Vercel, Docker, AWS (EC2, S3)
Machine Learning	Pandas, TensorFlow, Scikit-Learn, CNN, Transfer Learning

ACHIEVEMENTS

- **Amazon ML Summer School:** Selected among the top **3,000** out of **130,000+** applicants (~**2.3%** acceptance rate).
- **StealthFire Hackathon (3rd place / 25 teams):** Built a QR-based digital student ID system that solves lost-card recovery.